

IMO(28th) – 1987

First Day

1. Let S be a set of n elements. We denote the number of all permutations of S that have exactly k fixed points by $p_n(k)$. Prove that $\sum_{k=0}^n k p_n(k) = n!$.
2. The prolongation of the bisector AL ($L \in BC$) in the acute angled triangle ABC intersects the circumscribed circle at a point N . From point L to the sides AB and AC are drawn the perpendiculars LK and LM respectively. Prove that area of the triangle ABC is equal to the area of quadrilateral $AKNM$.
3. Suppose $x_1, x_2, \dots, x_n$ are real numbers with $x_1^2 + x_2^2 + \dots + x_n^2 = 1$. Prove that any integer $k > 1$ there are integers e_i not all 0 and with $|e_i| < k$ such that $|e_1 x_1 + e_2 x_2 + \dots + e_n x_n| \leq \frac{(k-1)\sqrt{n}}{k^n - 1}$.

Second Day

4. Does there exist a function $f : N \rightarrow N$, such that $f(f(n)) = n + 1987$ for every natural number n ?
5. Prove that for every natural number $n \geq 3$ it is possible to put n points in the Euclidean plane such that the distance between each pair of points is irrational and each three points determine a non-degenerate triangle with rational area.
6. Let $f(x) = x^2 + x + p$, for $p \in N$. Prove that if the numbers $f(0), f(1), \dots, f[\sqrt{p/3}]$ are primes, then all the numbers $f(0), f(1), \dots, f(p-2)$ are primes.

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